

TUF+ DSA Checklist — Master Problem Index

Complete Curated Directory with Direct Working Links to LeetCode & GeeksforGeeks

Total Curated Problems: **250**
Core Modules: **10 Sections**

Supported Platforms: **LeetCode & GFG**

Status: **Ready for Interactive Tracking**

ARRAYS & HASHING (21 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
1	Two Sum	Easy	LeetCode	Solve Problem →
2	3 Sum	Medium	LeetCode	Solve Problem →
3	4 Sum	Medium	LeetCode	Solve Problem →
4	Majority Element I	Easy	LeetCode	Solve Problem →
5	Majority Element II	Medium	LeetCode	Solve Problem →
6	Leaders in an Array	Easy	GeeksforGeeks	Solve Problem →
7	Rearrange Array Elements by Sign	Medium	LeetCode	Solve Problem →
8	Spiral Matrix	Medium	LeetCode	Solve Problem →
9	Pascal's Triangle I & II	Easy	LeetCode	Solve Problem →
10	Rotate Matrix by 90 Degrees	Medium	LeetCode	Solve Problem →
11	Sort an Array of 0s, 1s and 2s	Medium	LeetCode	Solve Problem →
12	Kadane's Algorithm (Max Subarray)	Medium	LeetCode	Solve Problem →
13	Next Permutation	Medium	LeetCode	Solve Problem →
14	Find Missing and Repeating Number	Medium	GeeksforGeeks	Solve Problem →
15	Count Inversions	Medium	GeeksforGeeks	Solve Problem →
16	Reverse Pairs	Hard	LeetCode	Solve Problem →
17	Maximum Product Subarray	Medium	LeetCode	Solve Problem →
18	Merge Sorted Arrays Without Extra Space	Medium	LeetCode	Solve Problem →
19	Longest Consecutive Sequence	Medium	LeetCode	Solve Problem →
20	Subarray Sum Equals K	Medium	LeetCode	Solve Problem →
21	Count Subarrays with Given XOR K	Medium	GeeksforGeeks	Solve Problem →

BINARY SEARCH (26 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
22	Binary Search (Search X in Sorted Array)	Easy	LeetCode	Solve Problem →
23	Implement Lower Bound	Easy	GeeksforGeeks	Solve Problem →
24	Implement Upper Bound	Easy	GeeksforGeeks	Solve Problem →

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
25	Search Insert Position	Easy	LeetCode	Solve Problem →
26	First and Last Occurrence in Sorted Array	Medium	LeetCode	Solve Problem →
27	Search in Rotated Sorted Array I	Medium	LeetCode	Solve Problem →
28	Search in Rotated Sorted Array II	Medium	LeetCode	Solve Problem →
29	Find Minimum in Rotated Sorted Array	Medium	LeetCode	Solve Problem →
30	Find How Many Times Array is Rotated	Easy	GeeksforGeeks	Solve Problem →
31	Single Element in Sorted Array	Medium	LeetCode	Solve Problem →
32	Find Square Root of a Number	Easy	LeetCode	Solve Problem →
33	Find Nth Root of a Number	Easy	GeeksforGeeks	Solve Problem →
34	Find the Smallest Divisor	Medium	LeetCode	Solve Problem →
35	Koko Eating Bananas	Medium	LeetCode	Solve Problem →
36	Minimum Days to Make M Bouquets	Medium	LeetCode	Solve Problem →
37	Aggressive Cows	Medium	GeeksforGeeks	Solve Problem →
38	Book Allocation Problem	Hard	GeeksforGeeks	Solve Problem →
39	Find Peak Element	Medium	LeetCode	Solve Problem →
40	Median of Two Sorted Arrays	Hard	LeetCode	Solve Problem →
41	Kth Element of Two Sorted Arrays	Medium	GeeksforGeeks	Solve Problem →
42	Minimize Max Distance to Gas Station	Hard	LeetCode	Solve Problem →
43	Split Array - Largest Sum	Hard	LeetCode	Solve Problem →
44	Search in a 2D Matrix I	Medium	LeetCode	Solve Problem →
45	Search in a 2D Matrix II	Medium	LeetCode	Solve Problem →
46	Find Peak Element II (2D Matrix)	Medium	LeetCode	Solve Problem →
47	Matrix Median	Hard	GeeksforGeeks	Solve Problem →

RECURSION & BACKTRACKING (15 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
48	Pow(x, n)	Medium	LeetCode	Solve Problem →
49	Generate Parentheses	Medium	LeetCode	Solve Problem →
50	Power Set / All Subsequences	Medium	LeetCode	Solve Problem →
51	Check Subsequence with Sum K	Medium	GeeksforGeeks	Solve Problem →
52	Combination Sum	Medium	LeetCode	Solve Problem →
53	Combination Sum II	Medium	LeetCode	Solve Problem →
54	Combination Sum III	Medium	LeetCode	Solve Problem →
55	Subsets II	Medium	LeetCode	Solve Problem →

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
56	Letter Combinations of a Phone Number	Medium	LeetCode	Solve Problem →
57	Palindrome Partitioning	Medium	LeetCode	Solve Problem →
58	Word Search	Medium	LeetCode	Solve Problem →
59	N-Queens	Hard	LeetCode	Solve Problem →
60	Rat in a Maze	Medium	GeeksforGeeks	Solve Problem →
61	M-Coloring Problem	Medium	GeeksforGeeks	Solve Problem →
62	Sudoku Solver	Hard	LeetCode	Solve Problem →

LINKED LIST (21 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
63	Add Two Numbers in Linked List	Medium	LeetCode	Solve Problem →
64	Segregate Odd and Even Nodes in LL	Medium	LeetCode	Solve Problem →
65	Sort a LL of 0s, 1s and 2s	Easy	GeeksforGeeks	Solve Problem →
66	Remove Nth Node from Back of LL	Medium	LeetCode	Solve Problem →
67	Reverse a Linked List	Easy	LeetCode	Solve Problem →
68	Add One to a Number Represented by LL	Medium	GeeksforGeeks	Solve Problem →
69	Find Middle of Linked List	Easy	LeetCode	Solve Problem →
70	Delete the Middle Node of LL	Medium	LeetCode	Solve Problem →
71	Check if LL is Palindrome	Easy	LeetCode	Solve Problem →
72	Intersection Point of Two LL	Easy	LeetCode	Solve Problem →
73	Detect a Loop in Linked List	Easy	LeetCode	Solve Problem →
74	Find Starting Point of Loop in LL	Medium	LeetCode	Solve Problem →
75	Find Length of Loop in LL	Easy	GeeksforGeeks	Solve Problem →
76	Reverse LL in Groups of Size K	Hard	LeetCode	Solve Problem →
77	Rotate a Linked List	Medium	LeetCode	Solve Problem →
78	Merge Two Sorted Lists	Easy	LeetCode	Solve Problem →
79	Flattening of a Linked List	Medium	GeeksforGeeks	Solve Problem →
80	Sort Linked List	Medium	LeetCode	Solve Problem →
81	Clone a LL with Random and Next Pointer	Medium	LeetCode	Solve Problem →
82	Delete All Occurrences of Key in DLL	Medium	GeeksforGeeks	Solve Problem →
83	Remove Duplicates from Sorted DLL	Easy	GeeksforGeeks	Solve Problem →

BIT MANIPULATION & GREEDY (17 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
84	Minimum Bit Flips to Convert Number	Easy	LeetCode	Solve Problem →
85	Single Number - I	Easy	LeetCode	Solve Problem →
86	Single Number - II	Medium	LeetCode	Solve Problem →
87	Single Number - III	Medium	LeetCode	Solve Problem →
88	Divide Two Numbers Without Multi/Div	Medium	LeetCode	Solve Problem →
89	XOR of Numbers in Given Range	Medium	GeeksforGeeks	Solve Problem →
90	Assign Cookies	Easy	LeetCode	Solve Problem →
91	Lemonade Change	Easy	LeetCode	Solve Problem →
92	Jump Game I	Medium	LeetCode	Solve Problem →
93	Shortest Job First (SJF)	Medium	GeeksforGeeks	Solve Problem →
94	Job Sequencing Problem	Medium	GeeksforGeeks	Solve Problem →
95	N Meetings in One Room	Easy	GeeksforGeeks	Solve Problem →
96	Non-overlapping Intervals	Medium	LeetCode	Solve Problem →
97	Insert Interval	Medium	LeetCode	Solve Problem →
98	Minimum Platforms for Railway	Medium	GeeksforGeeks	Solve Problem →
99	Valid Parenthesis Checker (with *)	Medium	LeetCode	Solve Problem →
100	Candy Distribution	Hard	LeetCode	Solve Problem →

SLIDING WINDOW & TWO POINTERS (10 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
101	Maximum Points You Can Obtain from Cards	Medium	LeetCode	Solve Problem →
102	Longest Substring Without Repeating Characters	Medium	LeetCode	Solve Problem →
103	Max Consecutive Ones III	Medium	LeetCode	Solve Problem →
104	Fruit Into Baskets	Medium	LeetCode	Solve Problem →
105	Longest Substring with At Most K Distinct Characters	Medium	GeeksforGeeks	Solve Problem →
106	Longest Repeating Character Replacement	Medium	LeetCode	Solve Problem →
107	Minimum Window Substring	Hard	LeetCode	Solve Problem →
108	Number of Substrings Containing All 3 Characters	Medium	LeetCode	Solve Problem →
109	Binary Subarrays with Sum K	Medium	LeetCode	Solve Problem →
110	Count Number of Nice Subarrays	Medium	LeetCode	Solve Problem →

STACKS & QUEUES (18 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
111	Implement Stack using Queues	Easy	LeetCode	Solve Problem →
112	Implement Queue using Stacks	Easy	LeetCode	Solve Problem →
113	Balanced Parentheses	Easy	LeetCode	Solve Problem →
114	Next Greater Element I	Easy	LeetCode	Solve Problem →
115	Next Greater Element II	Medium	LeetCode	Solve Problem →
116	Asteroid Collision	Medium	LeetCode	Solve Problem →
117	Sum of Subarray Minimums	Medium	LeetCode	Solve Problem →
118	Sum of Subarray Ranges	Medium	LeetCode	Solve Problem →
119	Remove K Digits	Medium	LeetCode	Solve Problem →
120	Implement Min Stack	Medium	LeetCode	Solve Problem →
121	Sliding Window Maximum	Hard	LeetCode	Solve Problem →
122	Trapping Rainwater	Hard	LeetCode	Solve Problem →
123	Largest Rectangle in Histogram	Hard	LeetCode	Solve Problem →
124	Maximal Rectangle	Hard	LeetCode	Solve Problem →
125	Stock Span Problem	Medium	LeetCode	Solve Problem →
126	The Celebrity Problem	Medium	GeeksforGeeks	Solve Problem →
127	LRU Cache	Medium	LeetCode	Solve Problem →
128	LFU Cache	Hard	LeetCode	Solve Problem →

TREES & BINARY SEARCH TREES (34 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
129	Level Order Traversal	Medium	LeetCode	Solve Problem →
130	Maximum Depth of Binary Tree	Easy	LeetCode	Solve Problem →
131	Check if Two Trees are Identical	Easy	LeetCode	Solve Problem →
132	Check for Balanced Binary Tree	Easy	LeetCode	Solve Problem →
133	Diameter of Binary Tree	Easy	LeetCode	Solve Problem →
134	Binary Tree Maximum Path Sum	Hard	LeetCode	Solve Problem →
135	Check for Symmetrical BT	Easy	LeetCode	Solve Problem →
136	Zig Zag or Spiral Traversal	Medium	LeetCode	Solve Problem →
137	Boundary Traversal of Binary Tree	Medium	GeeksforGeeks	Solve Problem →
138	Vertical Order Traversal	Hard	LeetCode	Solve Problem →
139	Top View of Binary Tree	Medium	GeeksforGeeks	Solve Problem →
140	Bottom View of Binary Tree	Medium	GeeksforGeeks	Solve Problem →

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
141	Left / Right View of Binary Tree	Medium	LeetCode	Solve Problem →
142	Root to Node Path in Binary Tree	Medium	GeeksforGeeks	Solve Problem →
143	Lowest Common Ancestor in BT	Medium	LeetCode	Solve Problem →
144	Maximum Width of Binary Tree	Medium	LeetCode	Solve Problem →
145	All Nodes Distance K in BT	Medium	LeetCode	Solve Problem →
146	Minimum Time to Burn the BT	Hard	GeeksforGeeks	Solve Problem →
147	Count Total Nodes in Complete BT	Medium	LeetCode	Solve Problem →
148	Construct BT from Preorder & Inorder	Medium	LeetCode	Solve Problem →
149	Construct BT from Postorder & Inorder	Medium	LeetCode	Solve Problem →
150	Serialize and De-serialize BT	Hard	LeetCode	Solve Problem →
151	Search in BST	Easy	LeetCode	Solve Problem →
152	Floor and Ceil in BST	Medium	GeeksforGeeks	Solve Problem →
153	Insert Node in BST	Medium	LeetCode	Solve Problem →
154	Delete Node in BST	Medium	LeetCode	Solve Problem →
155	Kth Smallest & Largest in BST	Medium	LeetCode	Solve Problem →
156	Validate Binary Search Tree	Medium	LeetCode	Solve Problem →
157	LCA in BST	Medium	LeetCode	Solve Problem →
158	Construct BST from Preorder	Medium	LeetCode	Solve Problem →
159	BST Iterator	Medium	LeetCode	Solve Problem →
160	Two Sum IV - Input is a BST	Easy	LeetCode	Solve Problem →
161	Recover BST (Nodes Swapped)	Medium	LeetCode	Solve Problem →
162	Largest BST in Binary Tree	Hard	GeeksforGeeks	Solve Problem →

HEAPS (5 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
163	Kth Largest Element in an Array	Medium	LeetCode	Solve Problem →
164	Kth Largest Element in a Stream	Easy	LeetCode	Solve Problem →
165	Heap Sort	Medium	GeeksforGeeks	Solve Problem →
166	Check if Array Represents Min Heap	Easy	GeeksforGeeks	Solve Problem →
167	Convert Min Heap to Max Heap	Medium	GeeksforGeeks	Solve Problem →

GRAPHS (32 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
168	Number of Provinces	Medium	LeetCode	Solve Problem →

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
169	Number of Islands	Medium	LeetCode	Solve Problem →
170	Flood Fill Algorithm	Easy	LeetCode	Solve Problem →
171	Rotting Oranges	Medium	LeetCode	Solve Problem →
172	Distance of Nearest Cell Having 1 (01 Matrix)	Medium	LeetCode	Solve Problem →
173	Surrounded Regions	Medium	LeetCode	Solve Problem →
174	Number of Enclaves	Medium	LeetCode	Solve Problem →
175	Detect Cycle in Undirected Graph	Medium	GeeksforGeeks	Solve Problem →
176	Bipartite Graph	Medium	LeetCode	Solve Problem →
177	Detect Cycle in Directed Graph	Medium	GeeksforGeeks	Solve Problem →
178	Topological Sort (Kahn's Algorithm)	Medium	GeeksforGeeks	Solve Problem →
179	Course Schedule I	Medium	LeetCode	Solve Problem →
180	Course Schedule II	Medium	LeetCode	Solve Problem →
181	Find Eventual Safe States	Medium	LeetCode	Solve Problem →
182	Alien Dictionary	Hard	GeeksforGeeks	Solve Problem →
183	Shortest Path in DAG	Medium	GeeksforGeeks	Solve Problem →
184	Word Ladder I	Hard	LeetCode	Solve Problem →
185	Word Ladder II	Hard	LeetCode	Solve Problem →
186	Dijkstra's Algorithm	Medium	GeeksforGeeks	Solve Problem →
187	Shortest Distance in a Binary Maze	Medium	LeetCode	Solve Problem →
188	Path With Minimum Effort	Medium	LeetCode	Solve Problem →
189	Cheapest Flights Within K Stops	Medium	LeetCode	Solve Problem →
190	Number of Ways to Arrive at Destination	Medium	LeetCode	Solve Problem →
191	Bellman Ford Algorithm	Medium	GeeksforGeeks	Solve Problem →
192	Floyd Warshall Algorithm	Medium	GeeksforGeeks	Solve Problem →
193	City With Smallest Number of Neighbors	Medium	LeetCode	Solve Problem →
194	Number of Operations to Make Network Connected	Medium	LeetCode	Solve Problem →
195	Accounts Merge	Medium	LeetCode	Solve Problem →
196	Making a Large Island	Hard	LeetCode	Solve Problem →
197	Most Stones Removed with Same Row or Column	Medium	LeetCode	Solve Problem →
198	Bridges in Graph (Critical Connections)	Hard	LeetCode	Solve Problem →
199	Articulation Point in Graph	Hard	GeeksforGeeks	Solve Problem →

DYNAMIC PROGRAMMING (39 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
200	Climbing Stairs	Easy	LeetCode	Solve Problem →
201	Frog Jump	Easy	GeeksforGeeks	Solve Problem →
202	Frog Jump with K Distances	Medium	GeeksforGeeks	Solve Problem →
203	Maximum Sum of Non-Adjacent Elements (House Robber)	Medium	LeetCode	Solve Problem →
204	Ninja's Training	Medium	GeeksforGeeks	Solve Problem →
205	Grid Unique Paths	Medium	LeetCode	Solve Problem →
206	Unique Paths II	Medium	LeetCode	Solve Problem →
207	Minimum Falling Path Sum	Medium	LeetCode	Solve Problem →
208	Triangle	Medium	LeetCode	Solve Problem →
209	Cherry Pickup II	Hard	LeetCode	Solve Problem →
210	Best Time to Buy and Sell Stock I	Easy	LeetCode	Solve Problem →
211	Best Time to Buy and Sell Stock II	Medium	LeetCode	Solve Problem →
212	Best Time to Buy and Sell Stock III	Hard	LeetCode	Solve Problem →
213	Best Time to Buy and Sell Stock IV	Hard	LeetCode	Solve Problem →
214	Best Time to Buy and Sell Stock with Fees	Medium	LeetCode	Solve Problem →
215	Subset Sum Equals to Target	Medium	GeeksforGeeks	Solve Problem →
216	Partition Equal Subset Sum	Medium	LeetCode	Solve Problem →
217	Partition Set into Two Subsets with Min Absolute Diff	Hard	GeeksforGeeks	Solve Problem →
218	0/1 Knapsack	Medium	GeeksforGeeks	Solve Problem →
219	Coin Change (Minimum Coins)	Medium	LeetCode	Solve Problem →
220	Target Sum	Medium	LeetCode	Solve Problem →
221	Coin Change II	Medium	LeetCode	Solve Problem →
222	Unbounded Knapsack / Rod Cutting	Medium	GeeksforGeeks	Solve Problem →
223	Longest Increasing Subsequence	Medium	LeetCode	Solve Problem →
224	Largest Divisible Subset	Medium	LeetCode	Solve Problem →
225	Longest String Chain	Medium	LeetCode	Solve Problem →
226	Longest Bitonic Subsequence	Medium	GeeksforGeeks	Solve Problem →
227	Number of Longest Increasing Subsequences	Medium	LeetCode	Solve Problem →
228	Longest Common Subsequence	Medium	LeetCode	Solve Problem →
229	Longest Palindromic Subsequence	Medium	LeetCode	Solve Problem →
230	Minimum Insertions to Make String Palindrome	Medium	LeetCode	Solve Problem →
231	Shortest Common Supersequence	Hard	LeetCode	Solve Problem →

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
232	Distinct Subsequences	Hard	LeetCode	Solve Problem →
233	Edit Distance	Medium	LeetCode	Solve Problem →
234	Wildcard Matching	Hard	LeetCode	Solve Problem →
235	Matrix Chain Multiplication	Hard	GeeksforGeeks	Solve Problem →
236	Minimum Cost to Cut the Stick	Hard	LeetCode	Solve Problem →
237	Burst Balloons	Hard	LeetCode	Solve Problem →
238	Palindrome Partitioning II	Hard	LeetCode	Solve Problem →

TRIES, ADVANCED STRINGS & MATH (12 PROBLEMS)

#	PROBLEM TITLE	DIFFICULTY	PLATFORM	DIRECT URL
239	Implement Trie (Prefix Tree)	Medium	LeetCode	Solve Problem →
240	Longest Word with All Prefixes	Medium	GeeksforGeeks	Solve Problem →
241	Number of Distinct Substrings in a String	Medium	GeeksforGeeks	Solve Problem →
242	Maximum XOR of Two Numbers in an Array	Medium	LeetCode	Solve Problem →
243	Maximum XOR With an Element From Array	Hard	LeetCode	Solve Problem →
244	Reverse Every Word in a String	Medium	LeetCode	Solve Problem →
245	Count and Say	Medium	LeetCode	Solve Problem →
246	Rabin Karp Algorithm	Medium	GeeksforGeeks	Solve Problem →
247	Shortest Palindrome	Hard	LeetCode	Solve Problem →
248	Longest Happy Prefix (KMP LPS)	Hard	LeetCode	Solve Problem →
249	Print All Primes till N (Sieve of Eratosthenes)	Medium	LeetCode	Solve Problem →
250	Prime Factorization of a Number	Easy	GeeksforGeeks	Solve Problem →